Detect Names from Legislative Speech Self-Introductions

Pallavi Das & Kasey Liu - CSC 482 W25 Group 4

Teammates

Pallavi Das

- Data Preprocessing
- Labeled Dataset
- Baseline NER & Regex Detection
- IRC Chatbot
- Finetuning spaCy Name Detection model
- Testing & Evaluation

Kasey Liu

- Labeled Dataset
- Training BERT Self-Intro Classifier
- IRC Chatbot
- Finetuning spaCy Name Detection model
- Testing & Evaluation

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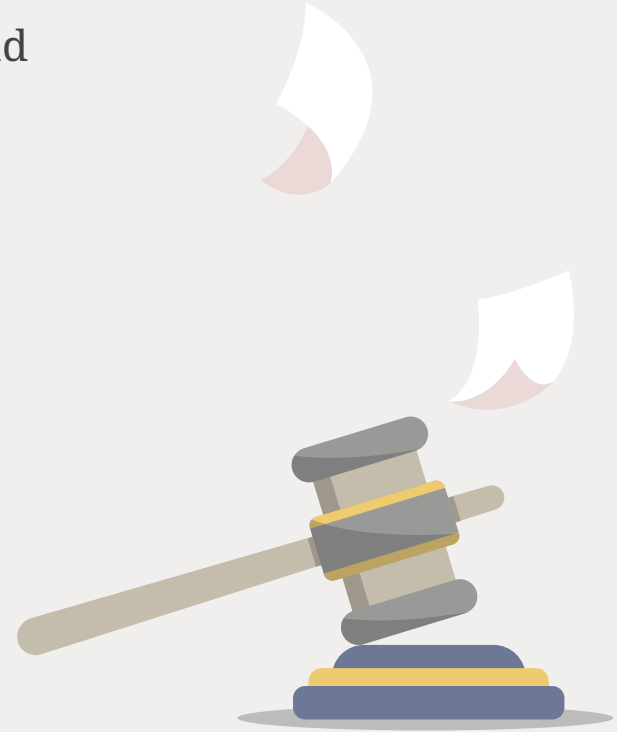
Introduction

Motivation & Problem

- Legislative hearing transcripts are hard to follow and not very accessible
- Generated using ASR leads to errors:
  - Misspelled names
  - Inconsistent formatting
- Major challenge is <u>speaker identification</u>
- No access to audio/video in many cases

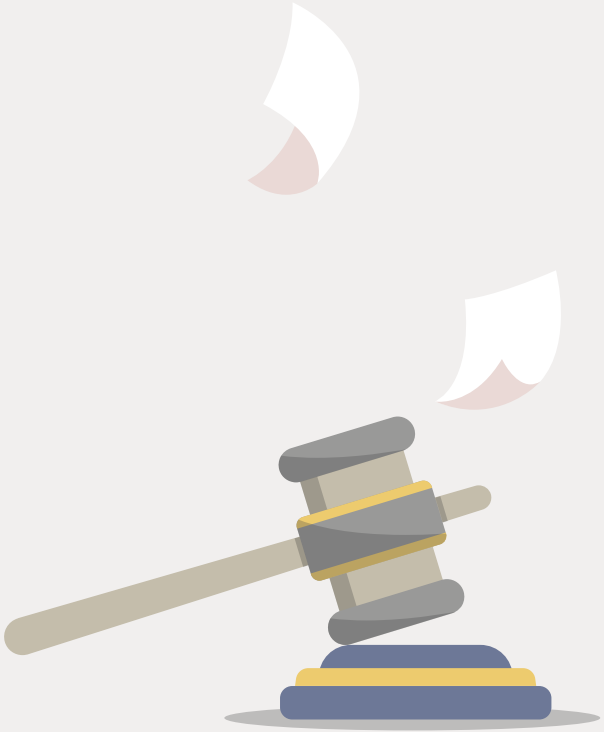


Why This is hard

- Many names mentioned aren’t the current speaker
- No explicit speaker labels
- Highly variable introduction formats
- Existing tools not designed for legislative data

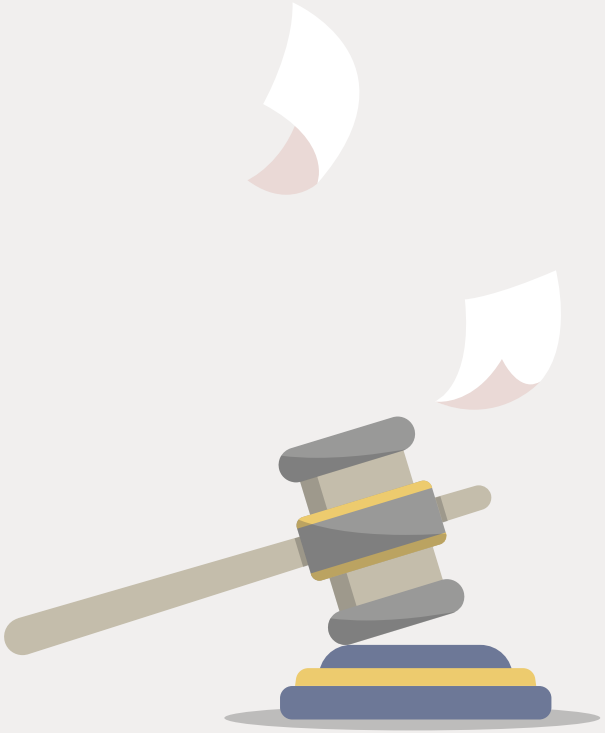
Example:

- "I'm Senator Khosmood"
- "My name is Foaad"
- "Foaad Khosmood, on behalf of Cal Poly"



Our Goal

- Detect all names in transcript text
- Identify if utterance is self-introduction
- Extract speaker name automatically



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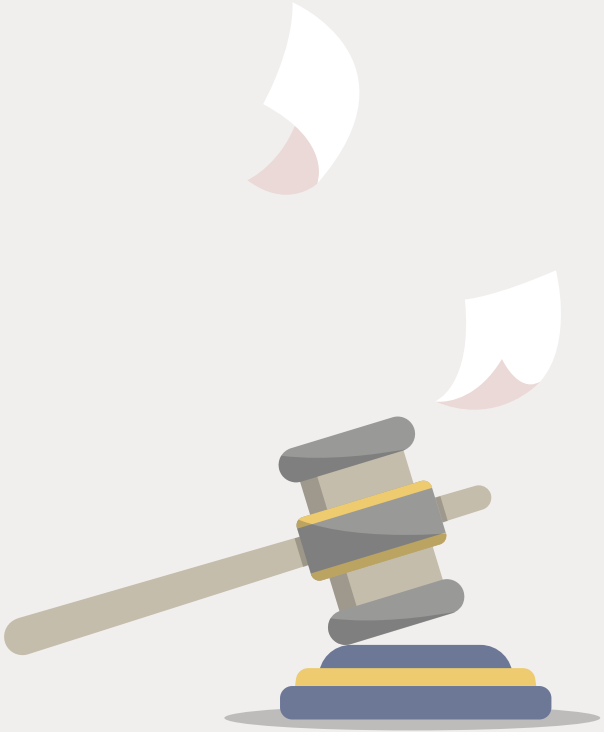
Implementation

Labeling the Dataset

- Source corpus: Digital Democracy California 2017-2018
  - Contained 392,565 utterances

- Created our own manually labeled dataset of 599 utterances
- Each labeled row includes:
  - all_names
  - is_self_intro
  - self_intro_name
  - true_speaker

- Label distribution:
  - 19% self-intros, 81% non-intros
  - 55% with names, 45% without names



sid	text	true_speaker	all_names	is_self_intro	self_intro_name
500189	Thank you Madam Speaker. As Chair of the Legislative Black Caucus said, it is a privilege we just had an opportunity to hear a great deal of the background on Frederick Roberts but it's our pleasure that we are able to recognize members of his family here today. The grandchildren of his own, his brother William Roberts Descendants of his family and with the help of the minority floor leader and the Freedmen Bureau we were able to make connections to the family and they are here with us today. And as part of the Robert's lineage they too are descendants of Sally Hemings and President Thomas Jefferson, the Author of the declaration of Independence. Robert's descendants are here on the floor with us and we'd like to recognize them now. Allen Wales Haddinett great niece of Frederick Roberts, Robin Johanna Roberts great niece of Frederick Roberts, and Karen Willard great, great niece of Frederick Roberts. Please join me in welcoming the descendents of Frederick Roberts.	Chris Holden	Frederick Roberts | William Roberts | Sally Hemings | Thomas Jefferson | Allen Wales Haddinett | Robin Johanna Roberts | Karen Willard	0	
327380	Wood? Here. Maienschein? Bonta? Burke? Gallagher? Here. Limon? Here. Mccarty? Nazarian? Present. Patterson? Here. Quirk-Silva? Here. Chiu? Here. Rodriguez? Here. Santiago? Here. Thurmond? Waldron? Here.	Committee Secretary	Wood | Maienschein | Bonta | Burke | Gallagher | Limon | Mccarty | Nazarian | Patterson | Quirk-Silva | Chiu | Rodriguez | Santiago | Thurmond | Waldron	0	
377808	Madam Chair and Members, Quick on the draw. Louie Brown on behalf of the California Association of Nurseries and Garden Centers, the California Strawberry Commission, and a few others listed in your analysis in support of the bill. And thanks to the author.	Louie Brown	Louie Brown	1	Louie Brown
556278	Yeah, so for example, this is a perfect example, your comments are well taken. I don't disagree with a single thing you just said. You're absolutely right.	Benjamin Allen		0	
506237	Chair and members, Jonathan Feldman, California Police Chiefs Association, in support.	Jonathan Feldman	Jonathan Feldman	1	Jonathan Feldman

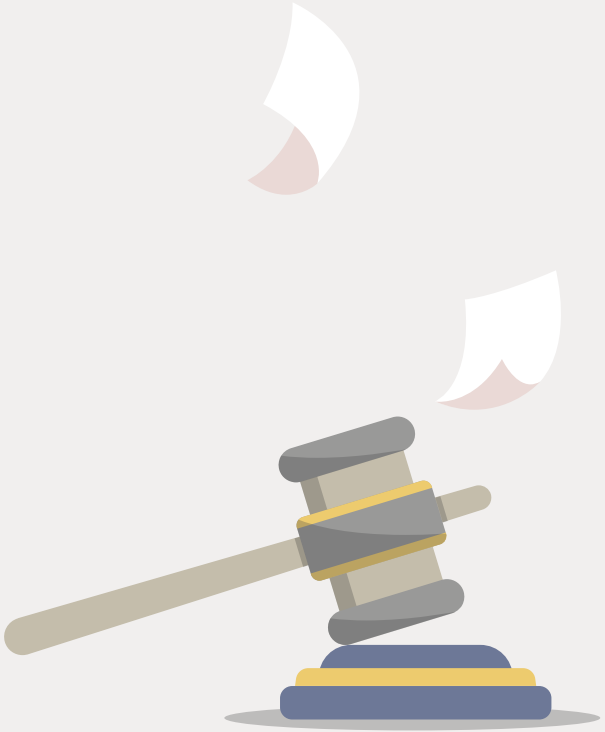
Self-Introduction Classifier (BERT)

- Initial approach: regex patterns
  - “My name is…”, “I am…”
- Problems:
  - Many false positives
    - "I am against this bill"
  - Missed non-standard introductions
- **Solution: BERT binary classifier**
  - Model: bert-base-uncased
  - Input: full utterance -> Output: self-intro (yes/no)
  - Trained on 599 labeled examples
  - Fine-tuned for 3 epochs (AdamW, PyTorch)
  - Model saved for real-time inference

Name Detection (spaCy NER)

- Goal: detect all person names in text
- Base model: spaCy transformer NER

- Converted label data to spaCy's character-offset format
- Froze non-essential components (tagger, parser, etc.)
- Trained NER head via custom Python loop for 20 epochs
- Handles:
  - partial names
  - multiple names per utterance
- Evaluated on CA 2015–2016 (unseen data)



System Pipeline w/ IRC Chat Bot Interface

Input: transcript utterance
classify "Hello I am Gavin and this is James, and we're with the state of California"

→

SpaCy NER - Extract Names
"Gavin" and "James"

→

BERT Classifier - Detect Self-Intro
Self-Intro Identified (1)

→

Combine Outputs - Predict Speaker
Gavin is speaker

Output:

- detected names
- self-intro prediction
- predicted speaker

"This is a self-introduction (confidence: 98.3%)! I believe the speaker is Gavin. Other names mentioned: James."

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Demo

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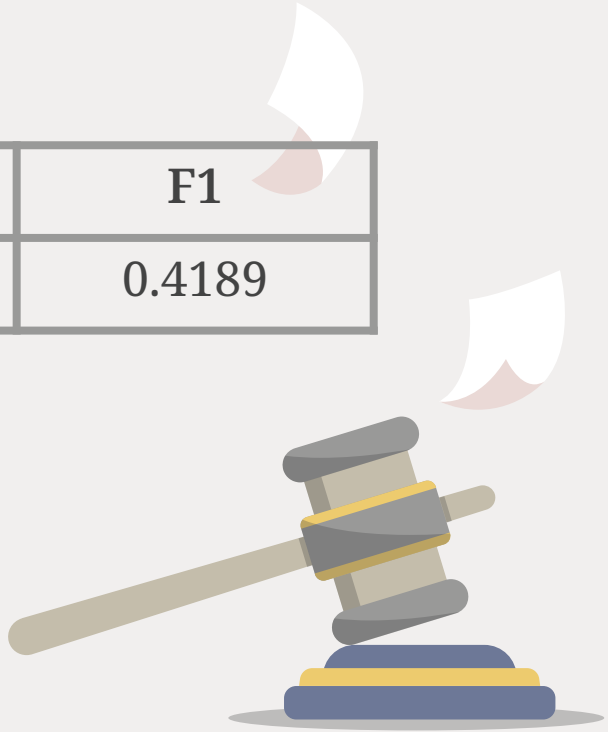
Results

Self-intro with Regex Patterns

Accuracy	Precision	Recall	F1
0.7129	0.3444	0.5345	0.4189

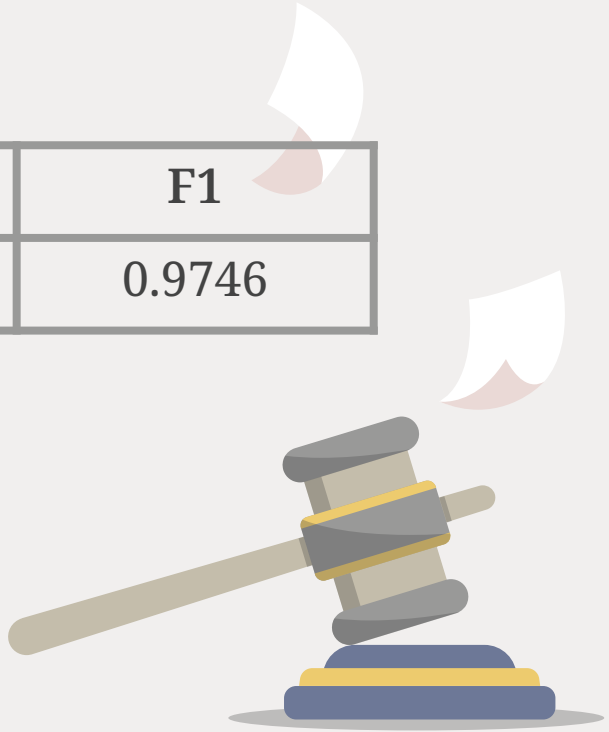
TN = 365	FP = 118
FN = 54	TP = 62



Self-intro with BERT Binary Classifier

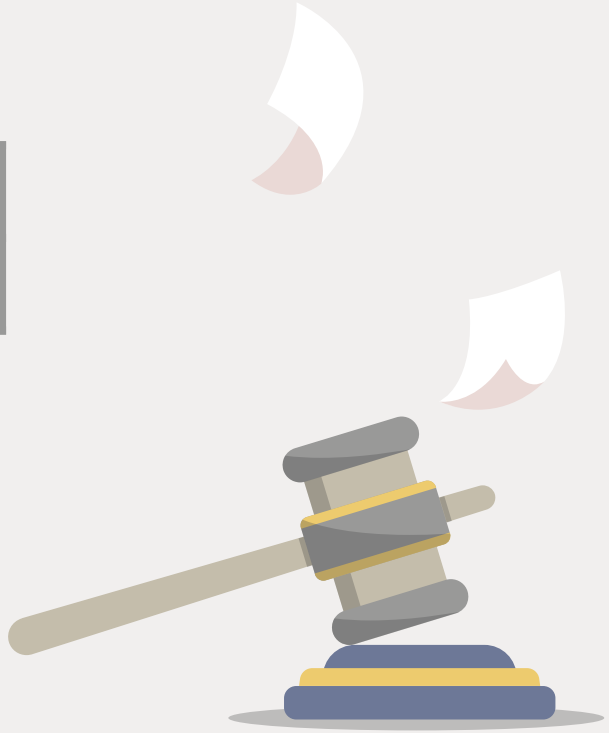
Accuracy	Precision	Recall	F1
0.9900	0.9583	0.9914	0.9746

TN = 478	FP = 5
FN = 1	TP = 115



Name Extraction with spaCy NER (Pretrained)

Precision	Recall	F1
0.8045	0.7226	0.7614



Name Extraction with spaCy NER (Trained on our dataset)

Precision	Recall	F1
0.9460	0.9520	0.9490

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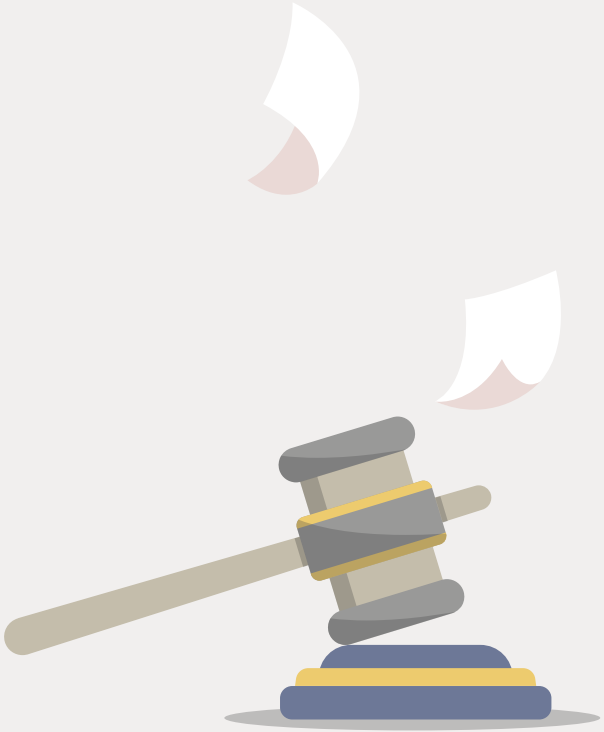
Conclusion

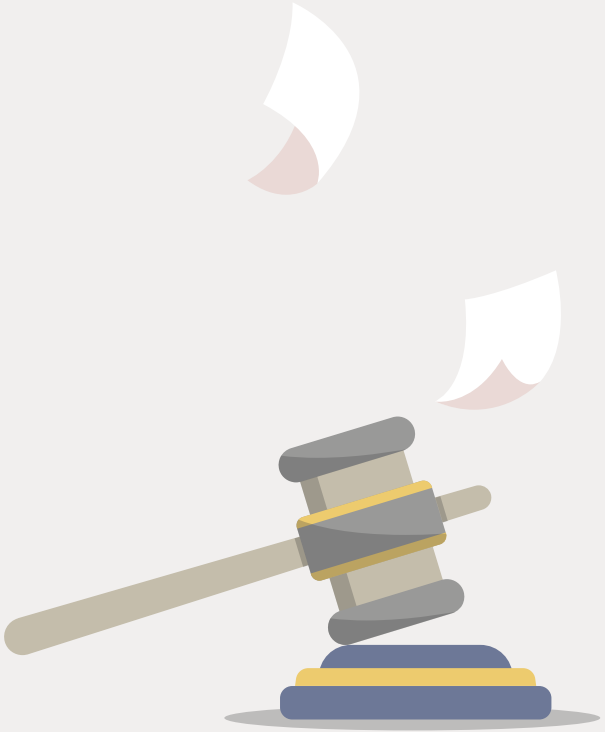
Conclusion

- Built specialized NER and speaker attribution system for legislative transcripts
  - using 599 manually labeled utterances
- Used a fine-tuned BERT classifier and a fine-tuned spaCy NER model
- Deployed system as IRC chatbot for interactive use



Future Work

- Expand training data across more states and sessions from the 1.6M+ speech corpus to improve generalization
- Look into other classifiers and models that may be better suited for our task
- Optimize pipeline for speed
- Build custom UI instead of using IRC



Thank you!

Questions?